

Timothy Medewase

medewaset@gmail.com | linkedin.com/in/timothy-medewase | github.com/TimothyMedewase | timothy-medewase.com

EDUCATION

The University of Texas at Arlington

Arlington, TX

M.S. in Computer Science, (Intelligent Systems/Robotics)

Aug. 2026 – May 2028

- Relevant Courses: Data Analysis and Modeling Techniques, Design and Analysis of Algorithms, Data Mining

University of Montevallo

Montevallo, AL

B.S. in Computer Science, Minor in Mathematics — GPA: 3.8/4.0

Aug. 2021 – May 2025

- Relevant Courses: Object-Oriented Programming, Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Calculus (I–III), Discrete Mathematics, Numerical Analysis, Complex Analysis, Data Analytics & Computing

EXPERIENCE

Software Engineer (AI/ML)

Jun. 2025 – Jun. 2026

Figbox Technologies, Inc

Upper Marlboro, MD

- Developed **ML-powered backend services** supporting **20,000+ active users** through data pipelines, inference workflows, and containerized deployments
- Built a natural-language search feature for user profiles by combining **spaCy**-based text processing with the **OpenAI API** for query understanding and ranking

Technologies Used: Python, JavaScript/TypeScript, Django, spaCy, React, Node.js, Docker, DigitalOcean

Software Development Intern

May 2023 – Jul. 2023

Solera, Inc

Westlake, TX

- Developed a **full-stack analytics dashboard** with data visualizations for internal reporting and analytics workflows
- Optimized **Microsoft SQL Server** stored procedures, reducing query latency by **50%**; integrated **RESTful APIs** with **Axios** and contributed to **CI/CD** workflows using **Bitbucket Pipelines**

Technologies Used: JavaScript, SQL, React, D3.js, Axios, Bitbucket Pipelines, Microsoft SQL Server

PROJECTS

Deep Research Agent | *Next.js, FastAPI, Python, Chroma, n8n, Supabase, LangChain*

Apr. 2026

- Built a research agent that decomposes queries into subquestions, runs iterative tool-calling loops, and streams cited answers over **SSE** under user-adjustable token and cost budgets
- Added **Chroma** session memory plus **n8n/Supabase** long-term recall, with context compression and multi-turn follow-ups that answer from memory or re-research

OpenBolts | *FastAPI, Next.js, React, PostgreSQL, Docker*

Jan. 2026

- Built an OEM knowledge layer for automotive repair with a hybrid **retrieval-augmented generation (RAG) pipeline** that ingests manufacturer manuals and delivers page-level citations with **sub-500ms average query response time**
- Implemented **hybrid retrieval scoring**, bounded refinement, context rewriting, and web fallback; developed real-time monitoring dashboards and per-phase ingestion tracking for reliable deployment at scale

SCOUR | *Swift, Supabase (PostgreSQL), FastAPI*

Oct. 2025

- Shipped an **iOS recruiting app** for athlete-coach matching, enabling athletes to browse and filter **15,000+ college programs** by sport, division, and location
- Integrated an **AI-assisted outreach and CRM backend** to manage subscriptions, track coach contacts, generate outreach emails, and synchronize **hundreds of athlete and coach records**

Transformer Implementation | *Python, PyTorch*

May 2025

- Implemented the **Transformer architecture** from the ground up in **PyTorch** with **6-layer encoder-decoder stacks**, **8 attention heads**, and **512-dimensional embeddings** following Vaswani et al
- Developed modular **multi-head attention**, residual connections, and sinusoidal positional embeddings; optimized attention computations for efficient training and **autoregressive decoding**

TECHNICAL SKILLS

Languages: Java, TypeScript/JavaScript, SQL, Python (Pandas, NumPy, TensorFlow, PyTorch, scikit-learn), C++, Swift, HTML/CSS

Frameworks/Technologies: React.js, Node.js, Express.js, FastAPI, Django, Spring Boot, LangChain, GitHub, Docker, AWS

Databases: PostgreSQL, MongoDB, Microsoft SQL Server, MySQL